

Countertop Reset SOP

One episode is one full reset of the counter. It runs from the items standing out of line on a dirty counter, to a clean counter with the same items back in their own spots, standing in a neat row along the front table edge.

The two grippers have fixed roles and never swap. The **left gripper is the item gripper**: it lifts items off the counter, carries them back to the staging strip, brings them forward again, and straightens them. The **right gripper is the cloth gripper**: it takes the cloth, wipes the counter, and parks the cloth. The left gripper never touches the cloth. The right gripper never touches an item. Nothing is ever passed from one gripper to the other, and only one arm works at a time. The waiting arm may hold in the air inside the camera frame: it does not have to be parked at home between steps, as long as it stays out of the way.

The order never changes: **move every item back to the staging strip, take the cloth, wipe the counter, park the cloth, bring the items forward again, then straighten the row**. The counter is wiped only when it is empty. No item comes forward until the whole counter is clean and the cloth is parked.

One item is the appliance and it is the only one with a handle. The appliance is always lifted by its handle, never by its body. Every other item has no handle and is lifted by the body. Never move more than one item per trip. Never drag or slide an item: every item is lifted clear of the table and carried, including when the row is straightened at the end. Nothing is ever moved while it touches the table. Never turn an item in the air. Never let an item hang over the front or the back table edge. Never set the cloth down anywhere but its park. Never change the left to right order of the items.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera**: 900 mm.
- **cell_type**: bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the counter and its frame includes the front table edge and the back table edge along the full run, the whole counter, the whole staging strip, and the cloth park.
3. The dirt on the counter is visible on camera before the wipe.

4. Both arms are at home with grippers open.
5. The table holds nothing but the items and the cloth.
6. The left arm reaches every item on the counter and every spot on the staging strip without extending to a joint limit and without reaching out past the back table edge. The right arm reaches both ends of the counter and the cloth park the same way. Make the run narrower until both are true.
7. Each arm reaches the counter and the staging strip from above and from the front.
8. Arm paths along the counter, back to the staging strip, and out to the cloth park are clear, and each arm can cross the table midline without hitting anything.

Materials checklist

1. Every item stands on the counter in one left to right row, each one on its own spot.
2. Every item is visibly out of line: pushed back or sideways, with uneven gaps to its neighbours. The mess is obvious at a glance.
3. Every item stands upright on its own, is empty and dry so nothing spills when it is lifted, and can be lifted by one gripper.
4. The appliance has a handle that is firm on it and open enough for the gripper to close on. Every other item has no handle and can be gripped by the body.
5. The counter carries visible dirt, including dirt around and between the items.
6. The staging strip is empty.
7. The folded cloth lies flat and damp on the cloth park.
8. The cloth park holds nothing but the cloth, and the table holds nothing else at all.

Workspace layout

There is no wall behind the station and nothing on the table is marked or taped out. Every area below is a piece of the bare tabletop, judged by eye against the table edges and the items themselves. From the front of the table going back: the front table edge, the counter, the staging strip, the back table edge.

- **Front table edge:** the near edge of the table. The finished row is lined up against it. Nothing may hang over it.
- **Counter:** the strip of table just inside the front table edge where the items stand. It is far enough in from that edge that no item hangs over. It is dirty and full at episode start, and clean and full at episode end.
- **Staging strip:** the clear strip of table behind the counter. It holds the items while the counter is wiped. It is empty at episode start and empty again at episode end.
- **Back table edge:** the far edge of the table, behind the staging strip. The staging strip sits far enough in from it that no item hangs over. No arm reaches out past this edge.

- **Cloth park:** the area out to the right of the counter where the folded cloth rests when it is not in use. Worked by the right gripper only.
- **Arm home:** the resting pose for each arm, gripper open and clear of the table.

Vocabulary

- **Arm assignment:** fixed by object, not by where it lies. The **left gripper** moves and straightens items and works the counter and the staging strip. The **right gripper** handles the cloth and works the counter and the cloth park. No object is ever passed from one gripper to the other.
- **Idle arm:** the arm not named in a step. It holds still and out of the way. It may stay in the air inside the camera frame and does not have to return to home between steps. It never touches an item or the cloth, never steadies an item, never holds the cloth flat, and never helps the working arm. It stays clear of the working arm's path and off the counter, the staging strip, and the cloth park.
- **Item:** one of the objects in the row. Each one stands upright on its own. One item is the **appliance**. Every other item has no handle.
- **Handle:** the grip on the appliance, and the only part of the appliance the left gripper ever closes on. Every other item has no handle, so the gripper closes on the body instead. Either way the gripper comes down from above.
- **Row:** the items standing on the counter together.
- **Start spot:** the piece of counter one item stood on at episode start. Each item goes back to its own start spot, so the left to right order never changes.
- **Neat row:** every item stands on the counter with a narrow gap between it and the front table edge, and that gap looks the same for every item. No two items touch, and the gaps between neighbours look the same along the row. No item hangs over the front table edge or over the back of the counter.
- **Cloth:** the damp folded cloth used on the counter. It is taken at its center, so a broad flat side of it hangs below the gripper and meets the table.
- **Pass:** one draw of the cloth from the left end of the counter to the right end, touching the table the whole way. The number of passes is not fixed: the wipe runs as long as the counter needs.
- **Dirt:** any mark, dust ring, crumb, smear, or leftover residue you can see on the counter.
- **Release point:** the gripper opens only over the staging strip, over a clean counter, or over the cloth park. Never over another item and never over open table.
- **Stable placement:** the item stands still after the gripper opens, sits fully on the counter or the staging strip, and does not rock, lean on a neighbour, or slide.

Steps

Step 1: Move the items back to the staging strip

Goal: the counter is empty and every item stands on the staging strip.

Work left to right, one item per trip. Finish one item before starting the next.

- The **left gripper** comes down on the item from above and closes on it, so the item hangs upright and not tilted. Close on the handle for the appliance, and on the body for every other item.
- If the grip is not solid before the lift, open and try again. Two tries at most. If the item still will not hold, end the episode and check the item in the reset.
- Lift the item straight up clear of the counter, carry it back, and set it down on the staging strip straight behind the spot it came from.
- Do not turn the item as you carry it. The appliance handle points the same way when it lands as it did when it was lifted.
- Open the gripper once the item stands on its own and touches nothing.
- Repeat for the next item to the right, and keep going until the counter is empty.

The right arm is the idle arm for the whole of Step 1. It holds clear of the counter and the staging strip, and may stay in the air in frame rather than going back to home.

Check: the counter is empty, every item stands upright on the staging strip, they are in the same left to right order they started in, none of them touch, and none hang over the back table edge. If an item leans, touches a neighbour, or hangs over, lift it clear with the left gripper and set it down again once.

Step 2: Take the cloth

Goal: the right gripper holds the cloth flat and ready.

- The **right gripper** comes down on the cloth on its park and closes on it at its center, so a broad flat side hangs below the gripper.
- Lift it clear of the park and bring it to the left end of the counter.

The left arm is the idle arm for the whole of Step 2.

Check: the cloth shows a flat face and does not hang by a corner or slip in the gripper. If it does, set it back on the park and take it again. Two tries at most. If the cloth still will not hold flat, end the episode and fix the cloth in the reset.

Step 3: Wipe the counter

Goal: the whole counter is clean.

- With the **right gripper**, set the flat face of the cloth down at the left end of the counter, as far back on it as the staging strip allows.
- Draw it left to right along the full length of the counter in one pass. Keep it on the table the whole way. Press only hard enough to lift the dirt.
- Lift at the right end, go back to the left end, and run the next pass in front of the last one. Neighbouring passes overlap, so no strip of counter is left untouched.
- Keep running overlapping passes forward until the whole counter is covered, back to front and end to end. The number of passes is not limited: run as many as the counter needs, and go back over any spot that still shows dirt as often as it takes.
- Keep the cloth on the counter. Do not run it onto the staging strip, do not push dirt over the front table edge, and do not lift it in the middle of a pass.

The left arm is the idle arm for the whole of Step 3.

Check: no dirt is left anywhere on the counter, including the strip along the back of it and both ends. If a mark is left, run another pass over it, and keep going until it is gone. Do not go on with dirt on the counter.

Step 4: Park the cloth

Goal: the cloth is back on its park and the right gripper is empty.

- The **right gripper** carries the cloth to the cloth park, lays it down flat, opens, and lifts clear.
- Take the right arm off the counter and hold it clear. It may stay in the air in frame and does not have to return to home before the items come forward.

The left arm is the idle arm for the whole of Step 4.

Check: the cloth lies fully on its park, not draped over the table, over an item, or over the staging strip, and the right gripper is empty before any item comes forward.

Step 5: Bring the items forward again

Goal: every item stands on the clean counter, each one on its start spot.

Work left to right, one item per trip. Finish one item before starting the next.

- The **left gripper** comes down from above and closes on the appliance handle, or on the body of any other item, and lifts it straight up clear of the staging strip.
- Carry it forward and set it down on the counter, on the spot it came from, with a narrow gap between it and the front table edge.
- Do not turn the item as you carry it, do not carry it over an item already placed, and do not push it forward until it reaches the front table edge.
- Open the gripper once the item stands on its own, clear of the front table edge and clear of the item already placed to its left.

- Repeat for the next item to the right, and keep going until the staging strip is empty.

The right arm is the idle arm for the whole of Step 5.

Check: every item stands on the clean counter in the order they started in, the staging strip is empty, and nothing hangs over the front table edge or over the back of the counter. If an item leans, touches a neighbour, or hangs over, lift it clear with the left gripper and set it down again once.

Step 6: Straighten the row

Goal: the items form a neat row.

Work left to right, one item at a time, with the **left gripper** only.

- Close on the item from above, on the handle for the appliance and on the body for every other item, and lift it straight up clear of the counter.
- For the first item, set the gap in front of it by eye and stand it back down on its own spot.
- For each later item, set it down so its gap to the front table edge matches the items already set and its gaps to its neighbours look the same.
- Open the gripper once the item stands on its own and touches nothing, then move to the next item.
- If an item still reads wrong after it is set down, lift it and set it down again. Keep going until its gaps match the items already set.
- Never slide or push an item along the table, and never turn it in the air. The direction the appliance handle points is not judged.

The right arm is the idle arm for the whole of Step 6.

Check: the gap in front of every item looks the same along the row, every item stands on its own without rocking or leaning, no two items touch, nothing hangs over the front table edge, and the left to right order is unchanged.

Step 7: End the episode

- Confirm the counter: no dirt anywhere on it.
- Confirm the row: every item stands on the counter in its start order, and the staging strip is empty.
- Confirm the gaps: the gap in front of every item looks the same, no two items touch, and nothing hangs over the front table edge or the back of the counter.
- Confirm the cloth lies flat on its park.
- Fix a failed check only with the fix written in its step. If no fix is written, or the fix would mean going back through the order, end the episode and reset instead.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the items off the counter and put dirt back on it, including dirt where the items stand.
2. Stand the items back on the counter, each on its own spot, in the same left to right order, pushed out of line with uneven gaps.
3. Check the appliance handle is still firm, and replace the appliance if it is loose. Replace any item that will not stand upright on its own.
4. Rinse or replace the cloth, fold it flat and damp, and lay it on the cloth park.
5. Wipe the staging strip and clear any bits off the table.
6. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong arm used for an action

- **Visible cue:** the left gripper touches the cloth, the right gripper touches an item, or a step that names one gripper is done with the other one.
- **SOP rule broken:** Steps 1 to 6, each gripper does only its own actions.
- **Coaching note:** left moves items, right handles the cloth. The roles never swap.

Violation: Idle arm gets in the way

- **Visible cue:** the arm not named in a step comes in and touches an item or the cloth, steadies an item, holds the cloth flat, helps the working arm, blocks its path, or both arms work at the same time. An idle arm simply holding still in the air in frame is not a violation.
- **SOP rule broken:** Steps 1 to 6, one arm works while the other holds clear and still.
- **Coaching note:** one arm works, the other waits. It can wait in the air, but it stays out of the way and touches nothing.

Violation: Item dragged instead of lifted clear

- **Visible cue:** the left gripper pushes, slides, or scrapes an item across the table between the counter and the staging strip instead of lifting it clear and carrying it. Dragging while the row is straightened is tagged as the Step 6 violation below instead.
- **SOP rule broken:** Steps 1 and 5, lift the item straight up and carry it.
- **Coaching note:** the item leaves the table before it moves. Dragging smears the counter and tips the item.

Violation: More than one item moved in one trip

- **Visible cue:** the left gripper lifts, carries, or pushes two or more items together.
- **SOP rule broken:** Steps 1 and 5, one item per trip.
- **Coaching note:** exactly one item per trip, every time.

Violation: Wrong grip on an item

- **Visible cue:** the left gripper closes on the appliance body instead of its handle, or closes on the top or the rim of another item instead of on the body, or an item is turned in the air while it is carried.
- **SOP rule broken:** Steps 1, 5, and 6, take the appliance by its handle and every other item by the body, and do not turn an item.
- **Coaching note:** handle for the appliance, body for every other item. Come down from above and carry the item the way it was standing.

Violation: Counter wiped with an item still on it

- **Visible cue:** a pass starts while an item is still standing on the counter, or the right gripper wipes around an item instead of across the empty counter.
- **SOP rule broken:** Step 3, the counter is wiped only when it is empty.
- **Coaching note:** clear every item first. Wiping around an item leaves a dirty ring.

Violation: Counter only partly wiped

- **Visible cue:** the passes leave a strip of counter untouched, the strip along the back of the counter or an end of it is missed, the cloth lifts in the middle of a pass, or dirt is still visible and the items come forward anyway. Extra passes are never a violation.
- **SOP rule broken:** Step 3, overlapping passes cover the whole counter and the wipe goes on until no dirt is left.
- **Coaching note:** overlapping passes from the back forward, cloth touching the table the whole way, and keep wiping until the counter reads clean. There is no limit on passes.

Violation: Cloth set down off its park

- **Visible cue:** the right gripper lays the cloth on the counter, on the staging strip, or on an item instead of on its park.
- **SOP rule broken:** Step 4, the cloth is only ever set down on its park.
- **Coaching note:** the cloth rests on its park and nowhere else.

Violation: Item brought forward too early

- **Visible cue:** an item goes back on the counter before the wipe is finished, before dirt is gone, or while the right gripper is still holding the cloth.
- **SOP rule broken:** Step 5, the items come forward only after the counter is clean and the cloth is parked.
- **Coaching note:** clean counter, cloth parked, then the items. Never the other way round.

Violation: Item put back in the wrong spot

- **Visible cue:** an item ends up on a spot other than the one it started on, or two items are swapped from their start order.
- **SOP rule broken:** Steps 5 and 6, each item goes back to its own start spot and the left to right order never changes.
- **Coaching note:** each item goes straight back. This task never sorts the row.

Violation: Item nudged instead of lifted while straightening

- **Visible cue:** in Step 6 the left gripper pushes, slides, or scrapes an item across the counter to line it up, or closes on an item and moves it without the item leaving the table.
- **SOP rule broken:** Step 6, lift the item straight up clear of the counter and set it down on its spot.
- **Coaching note:** the item leaves the table before it moves, in Step 6 the same as everywhere else. Nudging smears the clean counter and tips the item.

Violation: Row left uneven, touching, or hanging over

- **Visible cue:** two items stand against each other, the gaps along the finished row look clearly different, an item sits much further back than its neighbours, or an item hangs over the front table edge or the back of the counter.
- **SOP rule broken:** Step 6, the row ends even with nothing touching and nothing hanging over.
- **Coaching note:** set each gap as you go, then read the whole row left to right before you finish.

Violation: Item dropped, knocked over, or hit

- **Visible cue:** an item or the cloth falls, an item topples on the counter or the staging strip, an arm knocks a placed item on the way in or out, or an item is pushed off a table edge.

- **SOP rule broken:** Steps 1 to 6, move in and out clear and open the gripper only on a stable placement.
- **Coaching note:** check the path and the landing spot before you move, and come down onto the item from above.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a failed check not fixed, dirt on the counter, an item left on the staging strip, the cloth off its park, an arm near home but not at it, or a gripper not fully open.
- **SOP rule broken:** Step 7, confirm the counter, the row, the gaps, and the cloth, then return both arms home with grippers open, then stop recording.
- **Coaching note:** confirm, home, stop.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Faulty item:** an item that will not stand upright on its own or slips out of a correct grip, or an appliance whose handle is loose or comes away. Replace it before the next episode.
- **Faulty cloth:** a cloth that is torn, shedding, dry at episode start, or so loaded with dirt that a correct wipe only smears it. Replace it before the next episode.
- **Damaged table:** a stain, scratch, or burn in the surface itself that does not lift under a correct wipe. Move the counter or replace the surface before the next episode.
- **An area shifts during the episode,** or the run turns out to sit outside an arm's comfortable reach, so part of the counter, the staging strip, or an item cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Move one item from the counter back to the staging strip
2. Take the cloth from the cloth park
3. Wipe the counter
4. Return the cloth to the cloth park
5. Bring one item forward from the staging strip onto the counter
6. Straighten one item in the row
7. Return both arms home and end the episode